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Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Hannah Kettle MATHS TUTORIALS



Half Past Paper – Week 10

Higher Tier

Total Marks

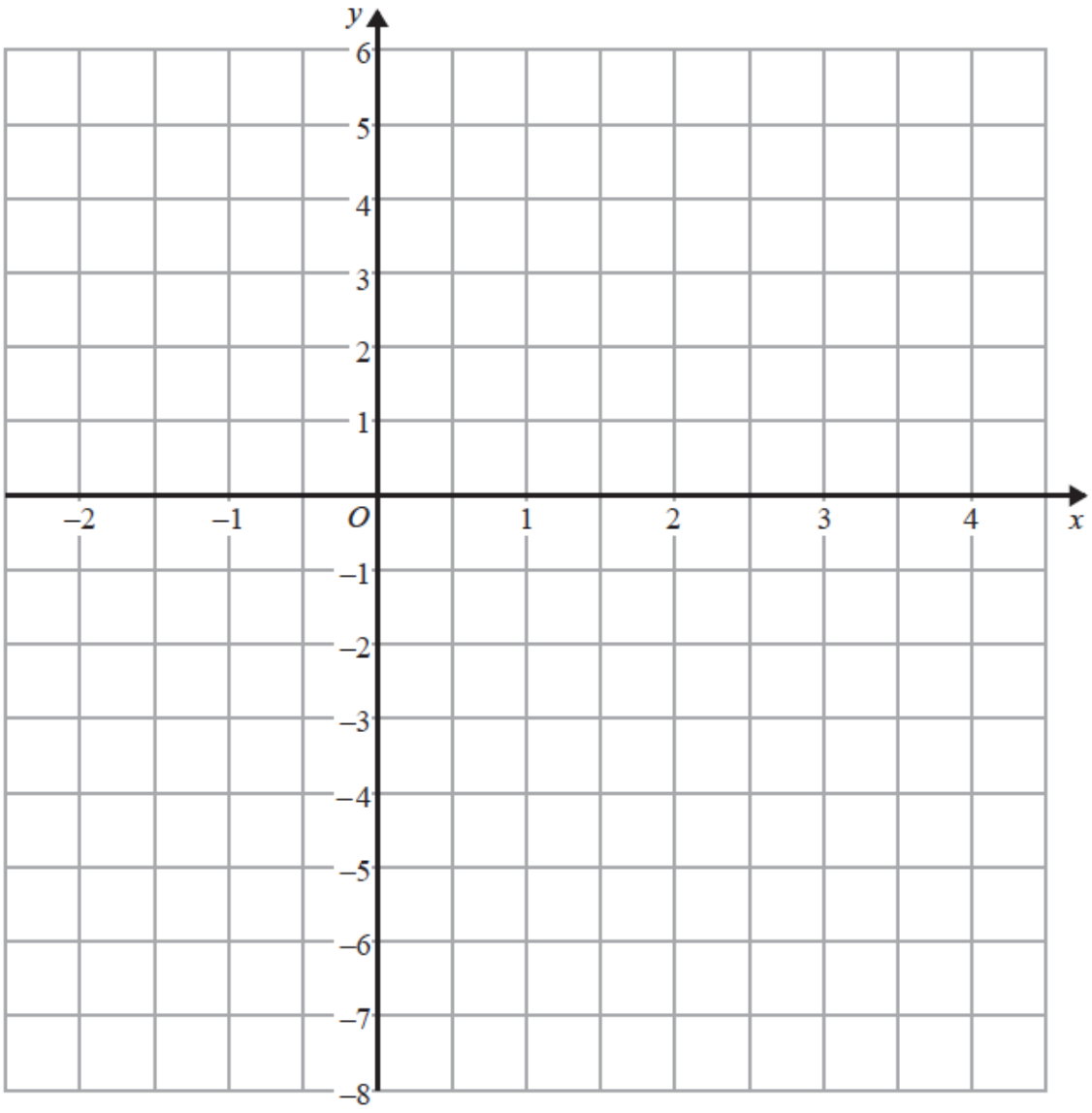
Q	Topic	Marks	Sparx Code
2	Plotting straight line graphs	3	U741
4	Finding the volume of cubes and cuboids	4	U786
6	Finding error intervals for truncated numbers	2	U301
8a	Using standard form with negative indices	1	U534
8b	Using standard form with positive indices	1	U330
10a	Tree diagrams for independent events	2	U558
10b	Tree diagrams for independent events	3	U558
12	Finding the area of sectors, Finding the arc length of sectors	4	U373,U221
14ai	Estimating gradients of non-linear graphs	3	U800
14aii	Calculating acceleration from velocity-time graphs	1	U562
14b	Estimating areas under non-linear graphs	3	U882
16	Equations of parallel and perpendicular lines	3	U898
18	Angles in segments and cyclic quadrilaterals	5	U251
20a	Solving geometric problems using vectors	2	U781
20b	Solving geometric problems using vectors	4	U781
	Total	41	

Answer ALL questions.

Write your answers in the spaces provided.

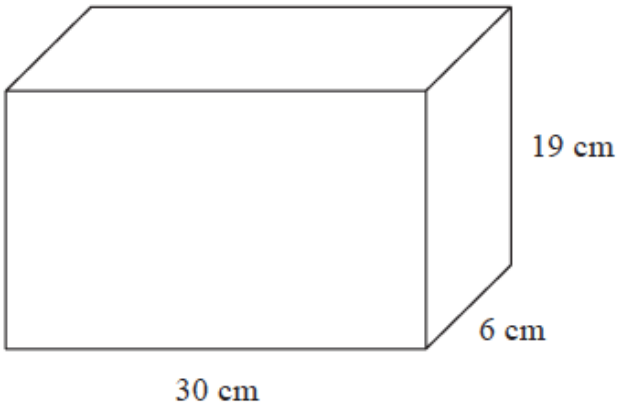
You must write down all the stages in your working.

2 On the grid below, draw the graph of $y = 2x - 3$ for values of x from -2 to 4



(Total for Question 2 is 3 marks)

4 A container is in the shape of a cuboid.



The container is $\frac{2}{3}$ full of water.

A cup holds 275 m*l* of water.

What is the greatest number of cups that can be completely filled with water from the container?

.....

(Total for Question 4 is 4 marks)

6 Sally used her calculator to work out the value of a number y.

The answer on her calculator display began

8.3

Complete the error interval for y.

..... $\leqslant y <$ .....

(Total for Question 6 is 2 marks)

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8 (*a*) Write 0.00562 in standard form.

.....

(1)

(*b*) Write 1.452×10^3 as an ordinary number.

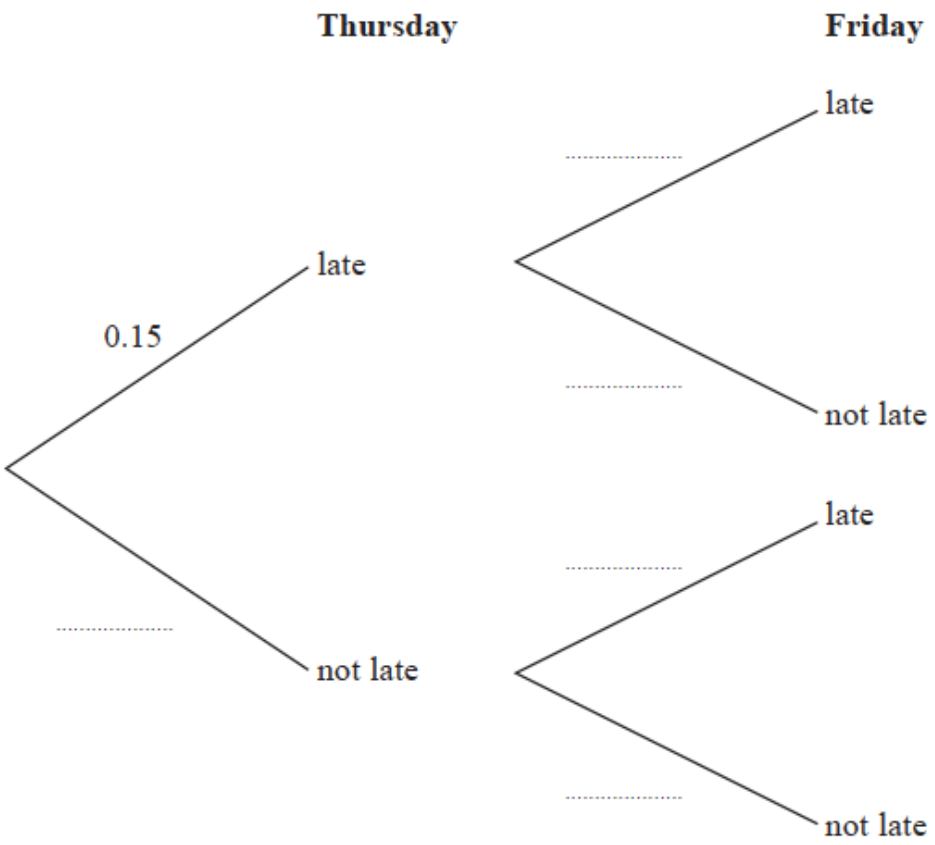
.....

(1)

(Total for Question 8 is 2 marks)

10 Mary travels to work by train every day.
The probability that her train will be late on any day is 0.15

(*a*) Complete the probability tree diagram for Thursday and Friday.



(2)

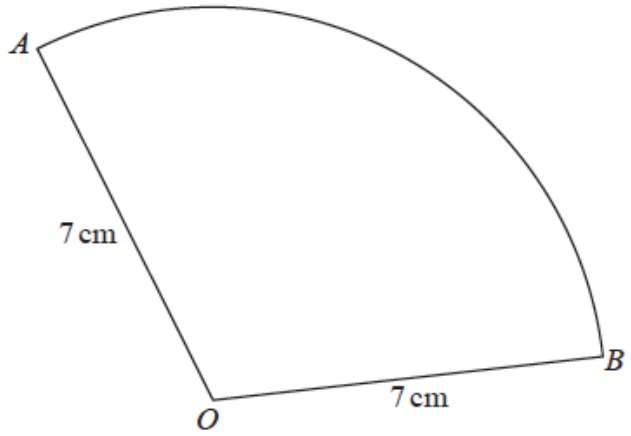
(*b*) Work out the probability that her train will be late on at least one of these two days.

.....

(3)

(Total for Question 10 is 5 marks)

12 *OAB* is a sector of a circle with centre *O* and radius 7 cm.



The area of the sector is 40 cm^2

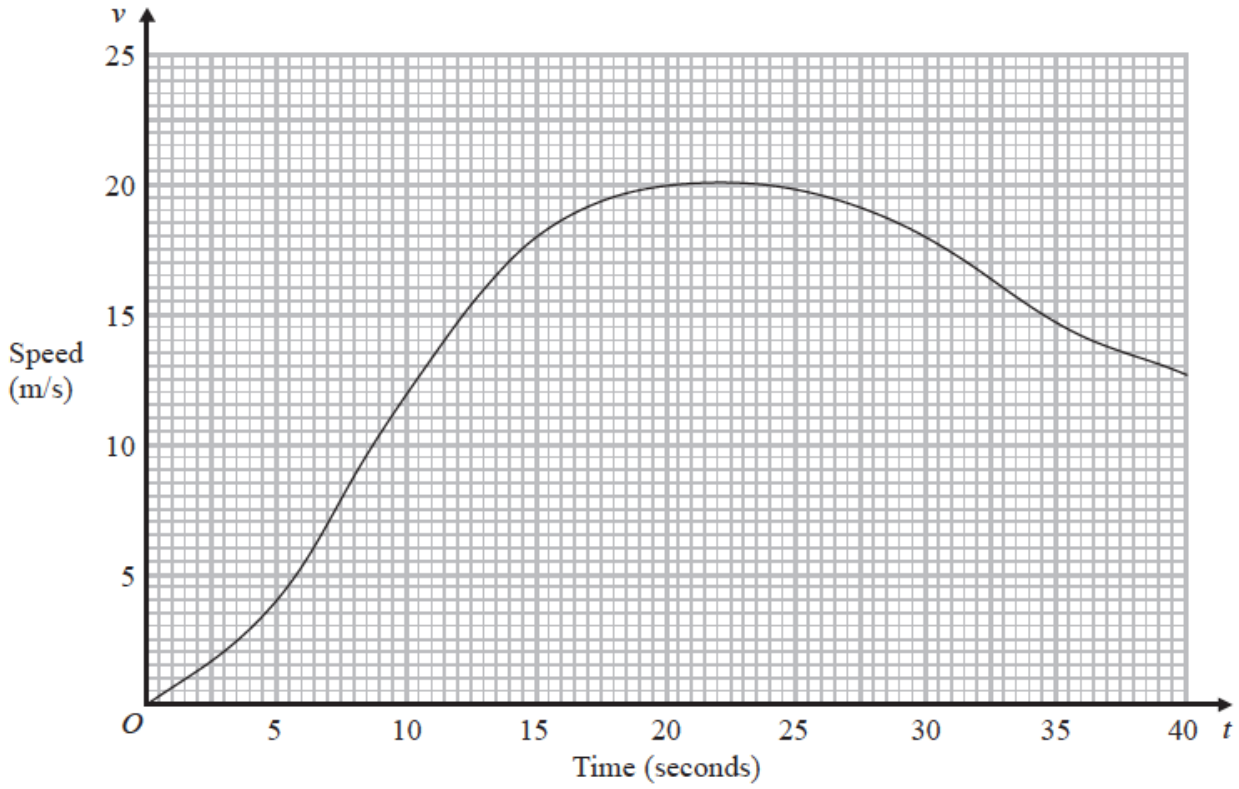
Calculate the perimeter of the sector.
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 12 is 4 marks)

14 A car moves from rest.

The graph gives information about the speed, v metres per second, of the car t seconds after it starts to move.



(*a*) (i) Calculate an estimate of the gradient of the graph at $t = 15$

.....

(3)

(ii) Describe what your answer to part (i) represents.

.....

(1)

(*b*) Work out an estimate for the distance the car travels in the first 20 seconds of its journey.
Use 4 strips of equal width.

.....m

(3)

(Total for Question 14 is 7 marks)

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16 The straight line **L** has the equation $3y = 4x + 7$
The point A has coordinates $(3, -5)$

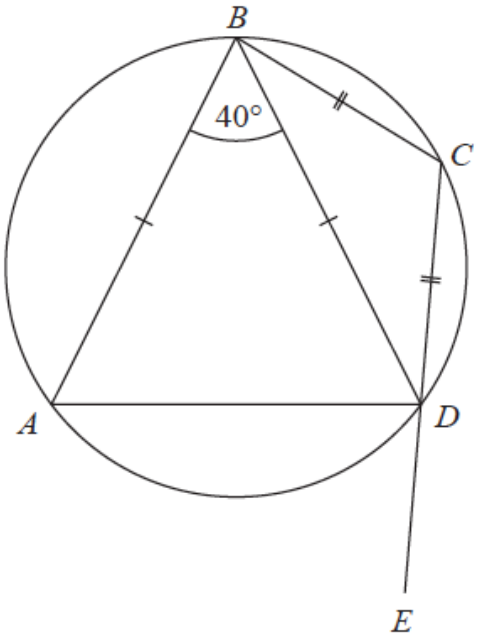
Find an equation of the straight line that is perpendicular to **L** and passes through A.

.....

(Total for Question 16 is 3 marks)

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18 The points *A*, *B*, *C* and *D* lie on a circle.
CDE is a straight line.

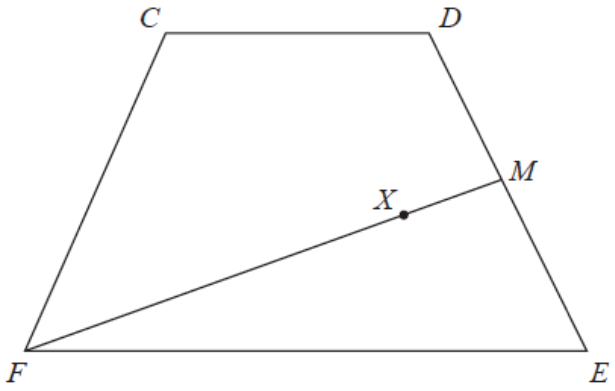


BA = *BD*
CB = *CD*
Angle *ABD* = 40°

Work out the size of angle *ADE*.
You must give a reason for each stage of your working.

(Total for Question 18 is 5 marks)

20 *CDEF* is a quadrilateral.



$\overrightarrow{CD} = \mathbf{a}$, $\overrightarrow{DE} = \mathbf{b}$ and $\overrightarrow{FC} = \mathbf{a} - \mathbf{b}$.

(*a*) Express $\overrightarrow{FE}$ in terms of **a** and/or **b**.
Give your answer in its simplest form.

.....

(2)

M is the midpoint of *DE*.
X is the point on *FM* such that *FX* : *XM* = *n* : 1
CXE is a straight line.

(*b*) Work out the value of *n*.

n = ...

(4)

(Total for Question 20 is 6 marks)

~~TOTAL FOR PAPER IS 41 MARKS~~